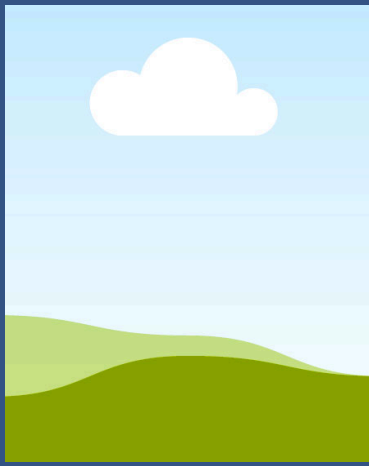




VO VAN TUAN

EMBEDDED/AUTOSAR INTERN EDUCATION

Vietnam – Korea University of Information and Communications Technology (VKU)

Faculty of Computer & Electronics Engineering
2021-2026(estimated)

WORK EXPERIENCE

IoT & Embedded System Developer (Student Projects)

Built HMI vehicle dashboard using Qt and ESP32, implemented real-time UART communication and hardware control.

Worked on FPGA-based systems (DE2 Cyclone II), including 7-segment display, switches, and clock divider logic. Basic knowledge of FreeRTOS task creation and timing. Practiced low-level peripheral interfacing and embedded driver code writing for UART, I2C, SPI.

Practiced Java/C++ backend programming: OOP, collections, and multi-threading.

PROJECT

Project: Smart Motorcycle Safety Monitoring System

- Developed a real-time motorcycle safety system using ESP32, integrating GPS, MPU6050, and GSM (SIM800L) for accident and fall detection.
- Implemented TinyML (Decision Tree/SVM) for on-device behavior classification.
- Built a Qt/QML GUI with Google Maps to visualize speed, location, and status.
- Simulated speed/fuel via potentiometers; communicated with PC via UART.
- Logged data locally (SPIFFS/MicroSD) and prepared for cloud upload (MQTT/Firebase).

Link: <https://github.com/TannFhongg/Smart-Motocycle-Safety-Monitoring-System>

Project: HMI-Car-ESP32

- Developed Qt-based HMI to visualize real-time vehicle data: speed, RPM, gear, ODO, and Trip.
- Interfaced ESP32 via UART (QtSerialPort), read analog signals from potentiometer, and digital signals from buttons.
- Used Q_PROPERTY and QqmlContext to bind C++ backend with QML UI.
- Persisted data using QSettings.
- Technologies: C++, QML, Qt, ESP32, UART, Git

Link: https://github.com/TannFhongg/ChuyenDe4_ProjectFinal

CONTACT INFORMATION

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SUMMARY

Final-year IoT student with hands-on experience in embedded systems, microcontrollers(ESP32), UART communication, and Qt-based HMI development. Looking for an embedded intern position in the automotive or industrial domain (e.g., AUTOSAR) to gain practical experience in firmware development, hardware interfacing, and real-time systems.

PERSONAL QUALIFICATIONS

Work skills:

- Strong in interested in existing technology and resolving problems by using it.
- Motivation to learn new technology.
- Having self-discipline and determination.

Professional skills:

- Programmin Languages: C/C++, Java, HTML/CSS
- Frameworks & Libraries: Qt
- Tools & Technologies: Visual Studio, Git
- Source control: Git
- Communication Protocols: UART, I2C, AT command set
- Storage: QSettings, SPIFFS, MicroSD
- Cloud Integration: MQTT, Firebase